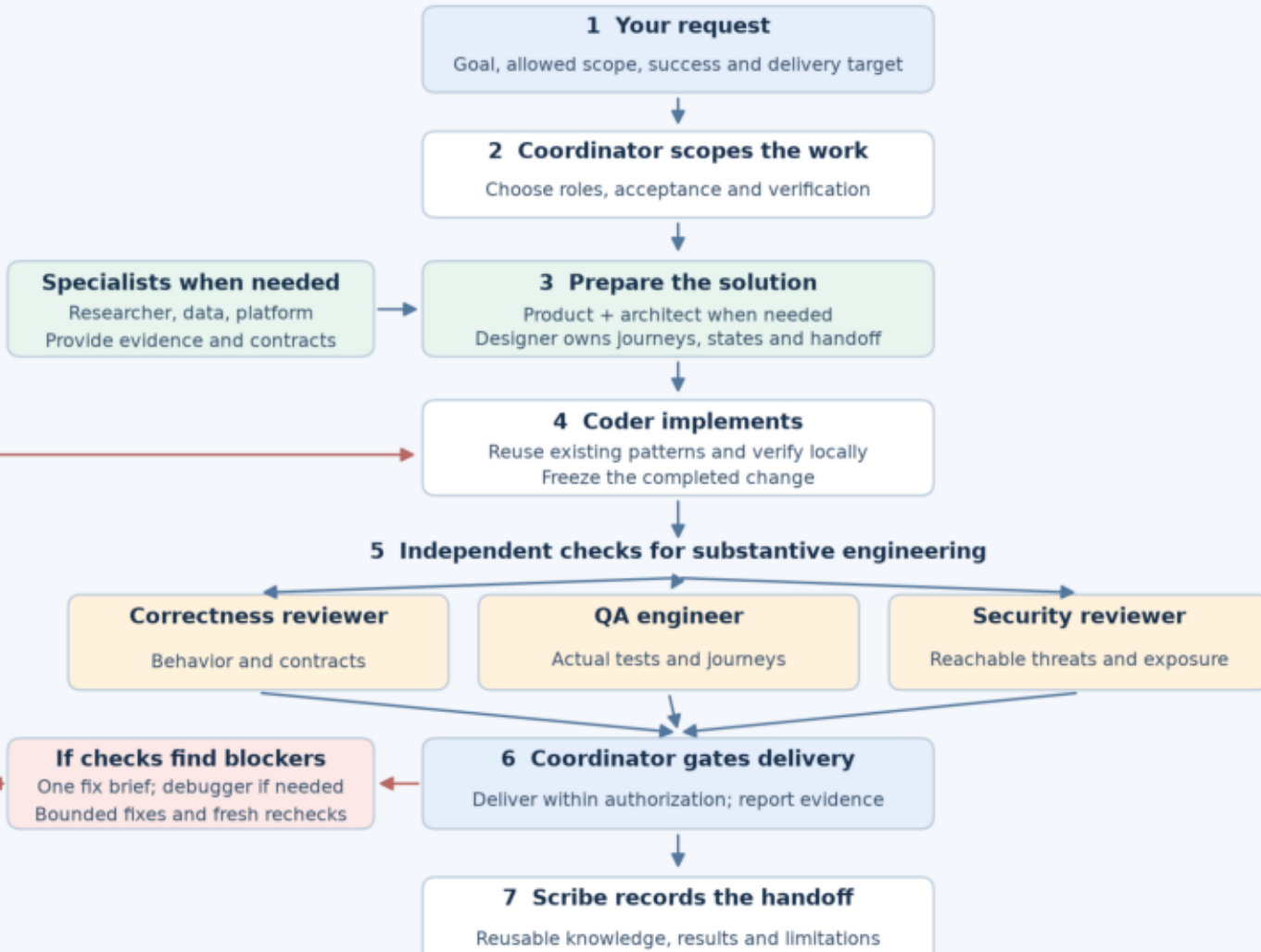
How the AI department works

One coordinator. Relevant specialists. Evidence before delivery.



Global skills and roles + project rules and permissions. Tools connect separately.

AI department | Skills catalog

Shared skills

ai-evaluation
Source: skills/ai-evaluation/SKILL.md

autonomous-development
Source: skills/autonomous-development/SKILL.md

commit-pr-standard
Source: skills/commit-pr-standard/SKILL.md

craftsman
Source: skills/craftsman/SKILL.md

design-workflow
Source: skills/design-workflow/SKILL.md

frank
Source: skills/frank/SKILL.md

research-workflow
Source: skills/research-workflow/SKILL.md

AI department | Setup and usage

How to use the AI department

Give your AI assistant a clear task. It uses shared skills to follow the team's methods and assigns specialist roles when needed. You remain responsible for business decisions and access; the coordinator manages the work and reports actual evidence.

Open the full workflow image (workflow.png) or editable vector (workflow.svg). See all twelve roles (../skills/frank/references/department.md).

Download the printable PDF guide (ai-department-guide.pdf) for setup, skills, department roles and resources.

Set up once

1. Clone this repository and open a terminal in its folder. Install Python 3.11 or newer and Git. Have a working Claude Code or Codex account.
2. Choose global setup to reuse the department across projects, or project setup for one repository. Replace the placeholders below with real paths; quote paths containing spaces.
3. Run the preview command. It lists proposed files without changing your setup. Then use the apply command when ready.

Global setup:

```
python scripts/install.py install --assistant both --scope user --target
"<home-directory>"
python scripts/install.py install --assistant both --scope user --target
"<home-directory>" --apply
```

One-project setup:

```
python scripts/install.py install --assistant both --scope project --target
"<project-root>"
python scripts/install.py install --assistant both --scope project --target
"<project-root>" --apply
```

Use --assistant codex or --assistant claude if you use only one assistant. Installation conflicts are refused. Read the backup and uninstall instructions (../README.md) before

replacing existing files.

Activate the guidance

The installer preserves your existing root instructions and account settings. Read the generated .claude/ai-department.md or .codex/ai-department.md inside the chosen target. Merge its shared guidance into the root instructions your assistant actually loads at that scope; keep existing instructions and resolve conflicts deliberately. Use your assistant's current instructions documentation to confirm its global instruction location.

Start a new session in your working project. Ask: "List the AI Department skills and role references you can actually access." Expect seven shared skills and twelve role definitions. Claude receives native agent definitions. Codex delegates native workers using the installed role references; these are not twelve automatically running agents. If assets are missing, inspect the install target and the assistant's discovery paths before assuming activation succeeded.

Keep common development methods global. Keep project commands, business rules, permissions and private knowledge in each project. Avoid duplicate project/global copies unless you deliberately manage which one takes precedence.

Ask for work

Copy an example and fill in the details:

Need - Example request
Build or fix - "Use autonomous-development to fix [problem] in [repository]. Success means [observable result]. Target is local. Preserve unrelated edits and verify in our development environment."
Design - "Use frank-designer and design-workflow to design [journey] for [users]. Reuse our components. Include mobile, loading, empty and error states, accessibility checks and a handoff for implementation."
Research - "Use frank-researcher and research-workflow to compare [tools]. Use dated primary sources. Consider our current stack, cost, latency, privacy and a measurable trial plan."
Evaluate AI - "Use ai-evaluation on [authorized held-out tasks]. Record actual outcomes, integrity checks, token/cost coverage and time. Separate synthetic demonstrations from live results."

Say which repositories may change, what success looks like and whether delivery means local files, a branch push or an authorized deployment. Missing tool connections must be reported. Installing this toolkit does not connect Figma, enable CI, buy models or provide credentials.

What happens next

The coordinator scopes your request. Relevant specialists prepare the brief, design or contracts; the coder implements. For substantive engineering, separate correctness, QA and security workers review one frozen version. Findings become one repair brief, followed by bounded rechecks. The coordinator delivers within your authorization and the scribe records reusable knowledge.

Simple questions and small design corrections do not need every department role. Ask for status if needed: "Show the current phase, active work versus waiting, completed checks and remaining blockers." A passing suite proves its tested cases, not universal AI accuracy or security.

AI department | Department roles

Department map

The main session coordinates the user's authorized outcome. These are reusable role instructions, not twelve permanently running processes or separate model subscriptions. Claude receives native agent definitions; Codex receives role references inside autonomous-development and delegates through native workers. Tool connections remain recipient-specific.

Role - Responsibility - Handoff
frank-product - Requirements, priorities, acceptance - Brief to designer/architect/coder
frank-designer - UX flows, prototypes, visual system, accessibility specification - Design/state matrix to coder and QA
frank-architect - System boundaries, contracts, migration decisions - Contracts and common vectors to builders
frank-coder - Frontend/backend implementation - Changes and focused evidence to reviewers
frank-researcher - AI/tool research and experiment design - Sourced decisions and evaluation plan
frank-data - Data models, retrieval, dataset provenance - Data contracts and evaluation inputs
frank-platform - Development environment, CI, operational observability - Reproducible runtime and release runbook
frank-reviewer - Independent correctness - Findings on frozen changes
frank-qa - Independent tests and user-journey verification - Actual results and coverage limits
frank-security - Independent threat and exposure review - Reachable security findings
frank-debugger - Root-cause diagnosis - Reproduction and repair recommendation
frank-scribe - Documentation and knowledge handoff - Maintained source-backed documentation

For a new interface: coordinator scopes the request; product clarifies acceptance if needed; designer maps journeys and states; coder implements; independent correctness, QA and security assess the frozen change. Architects, researchers, data and platform specialists join only where their responsibility is affected. Tiny layout fixes can use existing design patterns without a separate design task.

For AI research: researcher defines evidence and trial criteria; data prepares authorized inputs when needed; coder/platform build the bounded experiment; evaluation

records observed quality, latency and usage. Tool availability and account access must be verified, never assumed.

Each worker receives goal, files, acceptance, context, environment, evidence location and stopping condition. Workers do not recursively delegate or publish. Keep one coordinator, bounded parallelism and one combined fix brief. Standards and security claims require scoped evidence; no role can guarantee universal 100% reliability.

AI department | Reusable resources

Resources and tools

Skills provide decision instructions; resources provide reusable inputs and deterministic execution; connected tools provide access to models and external systems. Installing one does not establish the other two.

Capability - Bundled resource - Execution or connection
Engineering delivery - checkpoint contract (../skills/autonomous-development/references/checkpoint.md), six roles (../agents/frank-coder.md), craft standards (../skills/craftsman/SKILL.md) - Native assistant workers; Git; project's Docker/tests
Research - evidence ledger (../skills/research-workflow/references/evidence-ledger.md), primary-source catalog (research-sources.md) - Session web search/fetch or approved research connector
Design - journey/state acceptance matrix (../skills/design-workflow/references/acceptance-matrix.md) - Browser verification, project's components/assets; optional Figma
AI evaluation - schema and grading contract (../skills/ai-evaluation/references/evaluation.md), synthetic examples (../examples/evals/README.md), offline scorer - Recipient-owned model access plus independently graded held-out outcomes
Shared knowledge - global guidance (../templates/global-CLAUDE.md), project guidance (../templates/project-CLAUDE.md), scoped memory template (../templates/project-memory-MEMORY.md) - Fill with actual project commands and dated findings; no private memory copies
Tool verification - proof command (../scripts/prove.py), graphical report (verification/2026-10-07/report.md) - Docker; optional matplotlib only to redraw the chart

Research/design templates travel inside their installed skill folders. Scorer and role copies are generated from canonical sources. Project templates remain in this distribution for recipients to fill and merge deliberately.

Read the connection runbook (connections.md) for browser/model/design/source-control access and observed connection states. Missing API access, design libraries or real benchmark datasets cannot be replaced by a skill file. Do not treat a declared connection, sample dataset or mockup as observed external success.

AI department | Tools and trial priorities

AI development tooling: 07 October 2026

This shortlist combines current primary documentation with an independent Codex research challenge. It is a capability/release assessment, not a popularity ranking. No proposed integration below has been installed or connected by this research.

Direction and tool - Evidence - Decision for this environment
Focused skills and on-demand tool discovery - OpenAI's September guidance (https://developers.openai.com/blog/rethinking-skills-and-prompts-for-gpt-6-astra) recommends concise routing descriptions and loading supporting resources as needed. Anthropic tool discovery (https://www.anthropic.com/engineering/advanced-tool-use) defers large tool libraries. - Improve context routing first. Vendor token benchmarks do not establish our savings.
Browser verification with Playwright CLI and skills - Microsoft's Playwright MCP guidance (https://github.com/microsoft/playwright-mcp) recommends CLI plus skills for coding-agent efficiency while retaining MCP for richer persistent exploration. - Pilot against the already exercised MCP using identical journeys; compare tokens, time and successful assertions.
Agent tracing and evaluations with Langfuse - Official documentation (https://langfuse.com/docs) covers traces, cost/latency and experiments; self-hosting (https://langfuse.com/self-hosting) offers deployment options. - Limited tracing pilot to locate expensive retries and waiting. Local Compose is not a production HA/backup solution; define retention, redaction and access first.
Adversarial evaluation with Promptfoo - Current quickstart (https://www.promptfoo.dev/docs/red-team/quickstart/) covers injection/privacy testing and CI integration; configuration (https://www.promptfoo.dev/docs/red-team/configuration/) controls providers. - Pilot on authorized fixtures. Requires Node at least 22.22.0, newer than the observed local 22.20.0. Configure attack and grading providers explicitly; local execution does not imply local data processing.
Connected design context with Figma MCP - Figma documentation (https://developers.figma.com/docs/figma-mcp-server/) exposes components, variables, layouts and canvas workflows; approved-client restrictions apply. - Designer uses existing resources now. Pilot Figma only after verifying client eligibility and account access; keep private designs inside approved boundaries.
Background agents and event-driven workflows with Cursor - September releases (https://cursor.com/changelog) add project coordination and shipping tools. Bitbucket documentation (https://prod.cursor.com/docs/integrations/bitbucket) supports Cloud Agents and Bugbot for Bitbucket Cloud in public beta. - Controlled comparison, not a

runtime replacement. Account setup and actual behavior remain untested here; Rollouts compatibility is a separate question.

Evidence of broader movement exists: a June 2026 Codex usage study (https://arxiv.org/abs/2606.26959) reports active-user growth above fivefold during the first half of 2026. This describes that provider's populations, not market-wide tool rankings.

Adoption contract

Keep service-independent skills and resources canonical; each developer connects optional tools locally. Pin pilot versions, record environment and relevant inputs, and retain a rollback path. Avoid adding another orchestration framework unless an observed requirement needs it.

Compare baseline and candidate on the same held-out tasks with independent outcome grading. Record task success, critical/integrity failures, actual usage/cost coverage, active and waiting time, retries, and end-to-end latency including p95 when sample size permits. Missing measurements stay unknown. Inspect security findings and permission/data exposure before promotion. Reuse the ai-evaluation contract rather than counting retries as independent tasks.

The aspiration is complete accuracy and integrity for agreed tasks, with explicit speed budgets, scoped standards and security requirements. Finite trials cannot guarantee universal 100% accuracy or security. A passed test suite is evidence for those cases; standards compliance requires an identified standard, scope and assessment. No live-model accuracy, token reduction or production speed improvement has been established by this shortlist.